Nowcasting

Instructions

Workshop 2 – Workshop on Useful Statistical Procedure for Nowcasting Part 2: ARMA Models



IMF Institute for Capacity Development

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At the end of this workshop you should be able to:

- Estimate an ARMA(p,q) model;
- Work through the Box-Jenkins methodology for determining the appropriate values for p and q;
- Generate forecasts for an ARMA model.

The dataset for this workshop can be found in the page "Chile" of the work file **"arma_data.wf1**". We will be using the series CPI, which measures the price level of Chile for the period 1994M1 to 2009M12.

Question 1. Graph the data in the CPI series. Based on visual inspection, discuss the stationarity of the *cpi* series and any other salient feature that you notice.

Question 2. Calculate the ACF and PACF of the CPI series. What can you infer from the ACF/PACF about the structure of the ARMA model to predict CPI?

Question 3. Estimate an AR(1) model for the CPI series. Examine the stationarity of the estimated model by calculating the roots of the AR polynomial. Assess the presence of seasonality by examining the residuals visually. Calculate the ACF and PACF of the estimated model residuals.

Question 4. Calculate the 12 month inflation rate as p = ((cpi/cpi(-12))-1)*100. Inspect visually its graph. Is something happening around 2000? Is there a structural break? (Extra info: Chile introduced inflation targeting at the end of 1999.)

Question 5: Repeat Question 2 and 3 for inflation (i.e., the series labeled P) focusing on the post 2000 sample only. Experiment with an AR(1) and then an AR(2) model for P, and decide between these two models using the AIC-HQ-BIC statistics.

Question 6. Use the preferred model structure from question 5 with an extra MA(12) component, and estimate it again for the subsample (why?) 2000M1–2008M12. Use the estimated model to produce monthly forecasts for 2009-2010. Compare these forecasts with actual inflation for 2009, and determine the root mean square forecasting error.